

Nazmul Alam

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Dhaka, Bangladesh



Overview

Building robust web apps and AI systems using MERN, FastAPI, and PostgreSQL. Specialized in MLOps, orchestration and precision modeling.

Education

North South University

Bachelor of Science in Computer Science and Engineering

JAN 2015-DEC 2018

▶ CGPA : 3.62/4.0

Experience

Freelancing

MAY 2025 – PRESENT

Carboniq Project

REMOTE

- ▶ Built an AI-based carbon accounting platform with automated emissions tracking and scalable cloud architecture.
- ▶ Developed dashboards and reporting tools for sustainability analytics.

Lab Instructor

MAY 2019 – MAY 2025

North South University

DHAKA, BANGLADESH

- ▶ Conducted labs for C, Java, Databases, Data Structures, and Web Technologies; mentored students in programming and problem-solving.

Product Analyst

OCT 2022 – FEB 2024

Power Ledger Pty Ltd

PERTH, AUSTRALIA

- ▶ Supported project discovery, trading logic design, and technical handoffs; executed data analysis, reporting, and QA processes.

Sr. Technical Analyst

AUG 2021 – JUN 2023

InsideMaps Inc

DHAKA, BANGLADESH

- ▶ Analyzed operational data, automated trend reports, and collaborated with engineering teams to optimize workflows.

Jr. Data Scientist

NOV 2019 – AUG 2021

Me-Solshare Ltd

DHAKA, BANGLADESH

- ▶ Built predictive time-series models (CNN-based) achieving 95% accuracy to forecast energy consumption and support cost-saving decisions.

Intern

MAY 2019 – AUG 2019

Grameenphone Limited

DHAKA, BANGLADESH

- ▶ Supported BI teams in operational analytics and technical tasks.

Projects

CarbonIQ

MLOPS • WEB DEVELOPMENT

- ▶ Designed a **scalable carbon accounting platform** using **FastAPI, PyTorch, Airflow, PostgreSQL, Docker, Kafka**.
- ▶ Developed **time-series forecasting pipelines** to validate CO₂e estimates from consumption and real-time grid data.
- ▶ Built **automated ETL/ELT workflows** with Airflow/dbt and a **multi-tenant API architecture** with PostgreSQL for scalability and reliability.
- ▶ Optimized pipeline performance, reducing latency and improving model evaluation times.

EcoPulse

MLOPS • PYTORCH

- ▶ Developed an end-to-end **MLOps pipeline** using PyTorch, Airflow, dbt, MLflow, and FastAPI for reproducible training and deployment.
- ▶ Automated model **training, evaluation, versioning, and CI/CD** with GitHub Actions.
- ▶ Integrated MLflow Model Registry with FastAPI for **serving and experiment tracking**.
- ▶ Built **containerized workflows** with Docker and enhanced observability with structured logging and artifact tracking.

Skills

Programming Python, C, C++, JavaScript/TypeScript, PHP, Java

Databases SQL, MySQL, PostgreSQL, MongoDB, BigQuery

Frameworks Django, FastAPI, MERN

ML Frameworks TensorFlow, PyTorch

DevOps & Cloud Docker, Docker Compose, GitHub Actions, Airflow, MLflow, AWS (Basics), Linux, Shell Scripting

Analytics & Productivity Tableau, Looker Studio, Google Sheets, Excel, Colab, Earth Engine, Git, LaTeX, Markdown, VS Code, Word, Jira, Trello, Basecamp, Zoho

Publications

End-to-End OCR Using Synthetic Dataset Generation for Noisy Conditions

2020

Md. Shopon, **Nazmul Alam Diptu**, Nabeel Mohammed. *International Joint Conference on Computational Intelligence*, 2020. DOI: 10.1007/978-981-15-3607-6_41 ****Deep-learning OCR model** trained via ****synthetic dataset generation**** for improved robustness in ****low-quality images****.**

Population Estimation of Rohingya Refugees Using Artificial Neural Networks

2019

Nahian Ahmed, **Nazmul Alam Diptu**, Md. Sakil Khan Shadhin, Md. Abrar Fahim Jaki, Md. Ferdous Hasan, Md. Naimul Islam, Rashedur M. Rahman. *Vietnam Journal of Computer Science*, 2019. DOI: 10.1142/S2196888819500246 ****Artificial Neural Network (ANN)** model for population estimation, comparing ****satellite-derived**** and ****data-driven approaches****.**

Early Detection of Glaucoma Using Fuzzy Logic in Bangladesh Context

2018

Nazmul Alam Diptu, Md. Asif Khan, Sujit Debnath, Abdullah Al Imam, Kazi Asfaq Ahmed Ador, Rashedur M. Rahman. *International Conference on Intelligent Systems*, 2018. DOI: 10.1109/IS.2018.8710490 ****Fuzzy-logic decision model** implemented for early ****glaucoma detection**** in ****healthcare diagnostics****.**

Certifications

Deep Learning Specialization

2022

Coursera — *DeepLearning.AI*

- ▶ Built CNNs, RNNs, and sequence models; applied optimization and regularization.

Machine Learning Specialization

2021

Coursera — *Stanford / DeepLearning.AI*

- ▶ Supervised, unsupervised learning & ML engineering with hands-on projects.